THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 5	Repeat the largest-minus-smallest step.
2	**(b)** 24	List the sequence.
3	**(a)** 10	The second digit can be any of 10 digits.
4	**(b)** 165	Add the numbers.
5	**(c)** 24	Subtract.
6	**(a)** 4	Find the palindrome.
7	**(b)** 968	Reverse and add.
8	**(a)** 3	Continue the process.
9	**(b)** 7	Repeat the largest-minus-smallest step.
10	**(c)** 19	List the sequence.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Largest minus smallest each time; $1234 \to 3087 \to 8352 \to 6174$ (1); Steps $= 3$ (1).	1+1
12	Apply the rule (1); Terms: 15, maximum: 52 (1).	1+1
13	Digit sum $= 12 \times 5 = 60$ (1); Divisible by 3: yes; divisible by 9: no (1).	1+1
14	Largest minus smallest each time; $2002 \to 2178 \to 7443 \to 3996 \to 6264 \to 4176 \to 6174$ (1); Steps $= 6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Number 1551 $=$ 1551; $1551 \div 11 = 141$ exactly (1); Digit sum $= 12$ (1); Palindrome reads the same backwards (1).	1+1+1
16	Build the row; Row: 2, 3, 5, 8, 13, 21, 34, 55 (1); 8th number $= 55$ (1); Sum $= 141$ (1).	1+1+1
17	Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9752, smallest 2579 (1); Difference 7173 (1).	1+1+1
	OR: Number 2882 $=$ 2882; $2882 \div 11 = 262$ exactly (1); Digit sum $= 20$ (1); Palindrome reads the same backwards (1).	1+1+1
18	Build the row; Row: 1, 4, 5, 9, 14, 23, 37, 60 (1); 8th number $= 60$ (1); Sum $= 153$ (1).	1+1+1
	OR: Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 8631, smallest 1368 (1); Difference 7263 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Number 12321 (1); Digit sum 9 (1); Divisible by 9 (1); Smallest number to add: 0 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
	OR: Reverse-and-add repeatedly (1); Sums: 143, 484 (1); Steps $= 2$ (1); Palindrome 484 has 3 digits (odd) (2).	1+1+1+2
20	Number 24542 (1); Digit sum 17 (1); Not divisible by 9 (1); Smallest number to add: 1 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
	OR: Reverse-and-add repeatedly (1); Sums: 143, 484 (1); Steps $= 2$ (1); Palindrome 484 has 3 digits (odd) (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 10	The second digit can be any of 10 digits.
2	**(a)** 228	Add the numbers.
3	**(a)** 24	Subtract.
4	**(b)** 3	Find the palindrome.
5	**(b)** 827	Reverse and add.
6	**(a)** 3	Continue the process.
7	**(a)** 6	Repeat the largest-minus-smallest step.
8	**(b)** 18	List the sequence.
9	**(b)** 10	The second digit can be any of 10 digits.
10	**(c)** 255	Add the numbers.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Apply the rule (1); Terms: 21, maximum: 52 (1).	1+1
12	Digit sum $= 14 \times 7 = 98$ (1); Divisible by 3: no; divisible by 9: no (1).	1+1
13	Largest minus smallest each time; $4651 \to 5085 \to 7992 \to 7173 \to 6354 \to 3087 \to 8352 \to 6174$ (1); Steps $= 7$ (1).	1+1
14	Apply the rule (1); Terms: 11, maximum: 40 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Number 3663 $= 3663$; $3663 \div 11 = 333$ exactly (1); Digit sum $= 18$ (1); Palindrome reads the same backwards (1).	1+1+1
16	Build the row; Row: 3, 5, 8, 13, 21, 34, 55 (1); 7th number $= 55$ (1); Sum $= 139$ (1).	1+1+1
17	Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9654, smallest 4569 (1); Difference 5085 (1).	1+1+1
	OR: Number 4994 $= 4994$; $4994 \div 11 = 454$ exactly (1); Digit sum $= 26$ (1); Palindrome reads the same backwards (1).	1+1+1
18	Build the row; Row: 2, 6, 8, 14, 22, 36, 58, 94 (1); 8th number $= 94$ (1); Sum $= 240$ (1).	1+1+1
	OR: Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9821, smallest 1289 (1); Difference 8532 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Number 31713 (1); Digit sum 15 (1); Not divisible by 9 (1); Smallest number to add: 3 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
	OR: Reverse-and-add repeatedly (1); Sums: 143, 484 (1); Steps $= 2$ (1); Palindrome 484 has 3 digits (odd) (2).	1+1+1+2
20	Number 46264 (1); Digit sum 22 (1); Not divisible by 9 (1); Smallest number to add: 5 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
	OR: Reverse-and-add repeatedly (1); Sums: 143, 484 (1); Steps $= 2$ (1); Palindrome 484 has 3 digits (odd) (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 26	Subtract.
2	**(c)** 2	Find the palindrome.
3	**(a)** 1292	Reverse and add.
4	**(a)** 6	Continue the process.
5	**(c)** 3	Repeat the largest-minus-smallest step.
6	**(a)** 16	List the sequence.
7	**(a)** 510	Add the numbers.
8	**(d)** 18	Subtract.
9	**(b)** 5	Find the palindrome.
10	**(c)** 1352	Reverse and add.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Digit sum $= 11 \times 4 = 44$ (1); Divisible by 3: no; divisible by 9: no (1).	1+1
12	Largest minus smallest each time; $8532 \to 6174$ (1); Steps $= 1$ (1).	1+1
13	Apply the rule (1); Terms: 19, maximum: 160 (1).	1+1
14	Digit sum $= 15 \times 6 = 90$ (1); Divisible by 3: yes; divisible by 9: yes (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Number 5225 $=$ 5225; 5225$\div$11 $=$ 475 exactly (1); Digit sum $=$ 14 (1); Palindrome reads the same backwards (1).	1+1+1
16	Build the row; Row: 4, 7, 11, 18, 29, 47, 76 (1); 7th number $=$ 76 (1); Sum $=$ 192 (1).	1+1+1
17	Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9743, smallest 3479 (1); Difference 6264 (1).	1+1+1
	OR: Number 7447 $=$ 7447; 7447$\div$11 $=$ 677 exactly (1); Digit sum $=$ 22 (1); Palindrome reads the same backwards (1).	1+1+1
18	Build the row; Row: 1, 5, 6, 11, 17, 28, 45, 73, 118 (1); 9th number $=$ 118 (1); Sum $=$ 304 (1).	1+1+1
	OR: Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 8762, smallest 2678 (1); Difference 6084 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Number 50805 (1); Digit sum 18 (1); Divisible by 9 (1); Smallest number to add: 0 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
	OR: Reverse-and-add repeatedly (1); Sums: 726, 1353, 4884 (1); Steps $=$ 3 (1); Palindrome 4884 has 4 digits (even) (2).	1+1+1+2
20	Number 63936 (1); Digit sum 27 (1); Divisible by 9 (1); Smallest number to add: 0 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
	OR: Reverse-and-add repeatedly (1); Sums: 483, 867, 1635, 6996 (1); Steps $=$ 4 (1); Palindrome 6996 has 4 digits (even) (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.